

Integration on Manifolds

1.1 Integration of Differential Forms

Proposition 1.1.1

Suppose D and E are open domains of integration in $\mathbb{R}^n$ or $\mathbb{H}^n$, and $G: \bar{D} \rightarrow \bar{E}$ is a smooth map that restricts to an orientation-preserving or orientation-reversing diffeomorphism from D to E . If ω is an n -form on $\bar{E}$, then

$$\int_D G^* \omega = \begin{cases} \int_E \omega & \text{if } G \text{ is orientation-preserving,} \\ - \int_E \omega & \text{if } G \text{ is orientation-reversing.} \end{cases}$$

1.2 Stoke's Theorem

Theorem 1.2.1 ► Stokes's Theorem

Let M be an oriented smooth n -manifold with boundary, and let ω be a compactly supported smooth $(n-1)$ -form on M . Then

$$\int_M d\omega = \int_{\partial M} \omega. \quad (1.1)$$